Two routes to interpretability

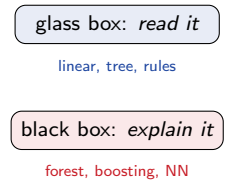
You built models that work — but *why* do they predict what they do?

Two routes:

- **Use a glass-box model you can read** — the model *is* the explanation. **(this deck)**
- **Explain a black box afterwards** with model-agnostic tools. (decks 2–3)

Today: the two transparent families you already know — **linear models** and **trees**.

Why bother? To **trust** a model, **debug** it, and catch ones that *cheat* — like deck 3's "wolf" detector that really keys on *snow*.



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- **Linear model:** yes — each feature has one coefficient.
- **A single tree:** yes — follow the splits from root to leaf (a rule).
- **A random forest: no** — hundreds of trees, no readable whole. But it still hands you a built-in *importance* number (which, we'll see, is biased).

Running example: **bike sharing** — predict daily rental count from weather & calendar features.

Outline

Reading a linear model

Reading trees

Rule-based models

So which features matter?

Reading a linear model

The most transparent model: one number per feature.

Coefficients as importance

A linear model $\hat{y} = \theta_0 + \sum_j \theta_j x_j$ tells you everything directly:

- **Sign** of θ_j = direction; **magnitude** = effect per $+1$ unit of x_j.
- **Standardize first** to compare — a raw coef of 5000 on `temp` (range 0–1) vs 2 on `hum` is *not* "2500× more important"; that's units (L12). Standardized, $|\theta_j|$ is fair: on bike `yr` & `atemp` top it.
- Classification analog: e^{θ_j} = **odds ratio** (L11).

No extra method needed: a (standardized) coefficient *is* a built-in importance and effect, in one number.

Linear models: the caveats

Only additive, linear effects

- One straight-line effect per feature; no interactions or curves unless you engineer them.

Correlated features split the credit

- `temp` and `atemp` (feels-like) are nearly the same; the model may give each half the coefficient — or unstable, even opposite, signs.

A coefficient answers "how does the prediction move with this feature, *holding the others fixed*" — which is exactly what breaks when features move *together*.

Lasso: the model selects its own features

L1 (Lasso) regularization shrinks coefficients toward zero — and drives many to **exactly zero** (callback: Ch. 2 regularization).

- Non-zero coefficients = the features the model *kept* — a built-in, sparsity-based **feature selection**.
- Stronger penalty ⇒ fewer survivors. **On bike (scaled):** the CV-tuned Lasso keeps 10/11; turn the penalty up and survivors drop $10 \to 4 \to 1$ — **sparsity is a knob you set**.

Caveat: among correlated features, Lasso keeps *one* and zeros the rest *arbitrarily* — "not selected" does not mean "unimportant."

Reading trees

One tree you can read; a forest you cannot — but it still scores features.

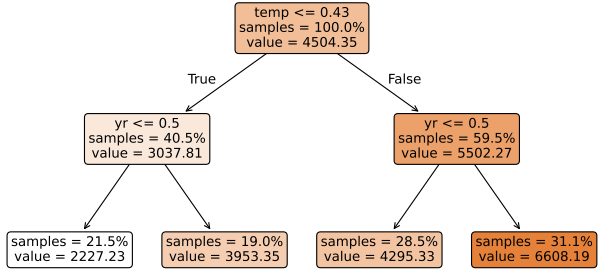
A single tree is a glass box

Each **root-to-leaf path is a rule**:

warm day (`temp`*>0.43) in year 2 (*`yr`*$=1$; 0=2011, 1=2012) $\rightarrow$ predict ~ 6608 rentals.*

- Fully transparent — if it stays shallow.
- Depth $\uparrow \Rightarrow$ accuracy $\uparrow$ but readability $\downarrow$.

A shallow regression tree on bike rentals (read a path as a rule)



How a split is chosen: reduce impurity

Recall from the trees lecture (ch. 4): a tree picks the split that most reduces **impurity** = parent − *weighted* child. Same idea, two rulers — one worked number each:

Classification: information gain (entropy H)

Toy: parent 5/5 $\Rightarrow H = 1.0$; a 4/1 child has $H = 0.72$.

$$\text{gain} = 1.0 - \underbrace{0.72}_{\text{weighted child}} = \mathbf{0.28}$$

The split with the biggest reduction wins.

Regression: variance reduction (bike root)

Split `temp` ≤ 0.43 ($n = 731$):

$$\text{reduction} = \underbrace{3.75\text{M}}_{\text{parent}} - \underbrace{2.32\text{M}}_{\text{weighted child}} \approx \mathbf{1.43}\text{M}$$

($\approx 38\%$ of the parent variance)

Forests: unreadable, but they score features

A random forest is **hundreds of trees** — no readable whole. Yet it gives a built-in **feature importance**:

> Sum each feature's impurity reduction (info gain / variance reduction) over *all* its splits, averaged across the trees ⇒ one importance per feature.

But it is biased:

- Favors **high-cardinality / continuous** features (more places to split).
- **Dilutes** across correlated features (credit shared between `temp`, `atemp`).
- Computed on **training** data ⇒ can reward overfitting.

Reading these in sklearn

```
# linear: standardized coefficients as importance
from sklearn.preprocessing import StandardScaler
from sklearn.linear_model import LinearRegression, LassoCV
from sklearn.pipeline import make_pipeline
Xs = StandardScaler().fit_transform(X)
coef = LinearRegression().fit(Xs, y).coef_           # |coef| =
    importance
sel  = make_pipeline(StandardScaler(), LassoCV())    # SCALE, then
    L1 (Lasso is scale-sensitive)
kept = sel.fit(X, y)[-1].coef_ != 0                  # features
    Lasso keeps (alpha tuned by CV)

# trees: built-in (impurity) importance + read one tree
from sklearn.ensemble import RandomForestRegressor
from sklearn.tree import DecisionTreeRegressor, plot_tree
RandomForestRegressor().fit(X, y).feature_importances_
plot_tree(DecisionTreeRegressor(max_depth=3).fit(X, y))
```

Rule-based models

Rules you can read — a tree's logic with a linear model's sparsity.

RuleFit: turn rules into features

A single tree is readable but brittle; a forest is accurate but unreadable. **RuleFit** goes for both, in three steps:

1. **Mine rules** from a tree ensemble: every root-to-node path becomes a binary rule — e.g. `(temp>0.35 and yr=1)`, which is 1 when the row falls in that region and 0 otherwise.
2. **Fit a sparse linear model** (Lasso) over *both* the original features and all these new rule-features.
3. **Keep the non-zero terms** — a short, weighted list of rules (plus a few surviving linear terms).

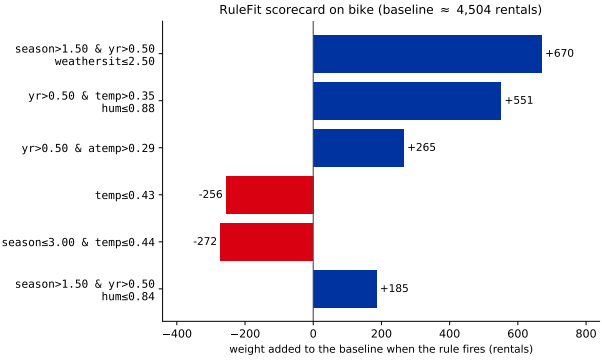
RuleFit = **rules from trees** + **Lasso selection** — the two glass boxes from this deck, combined into one readable model.

Reading a RuleFit model: a scorecard

Read it like a **scorecard**: start at the baseline ($\sim$4500 rentals, the average), then **add the weight of every rule that fires** for this day.

A warm, fair 2012 day fires the big **positive** rules (+670, +551, . . .) and stacks well above average; a cold day (`temp`$\leq$ 0.43) fires the **negative** ones. This 13-rule fit scores $R^2 \approx 0.65$.

Watch out: more rules $\Rightarrow$ more accurate but less readable (a denser RuleFit climbs toward the forest's $\sim$ 0.87); overlapping rules **share credit**, so one weight is not a feature's full effect.



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Same bike data, same goal. **Predict:** will the linear model and the random forest pick the *same* top features?

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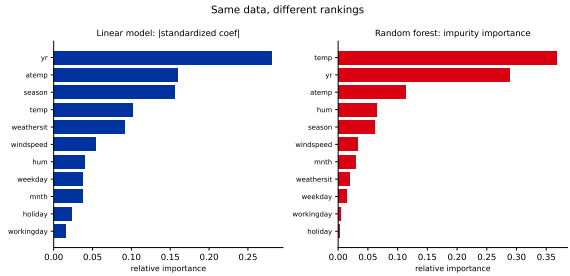
They don't:

- Linear: `yr`, `atemp`, `season` on top.
- Forest: `temp`, `yr`, `atemp` on top.

Part of the gap is the **impurity bias** we just saw; the rest is that they genuinely **measure different things** (a linear effect vs. impurity reduction), splitting the `temp`/`atemp` correlation differently.



Importance is method-dependent — always say *which* importance you mean.

Recap

- **Linear models** are read directly: (standardized) **coefficients** = effect + importance; e^{θ_j} = odds ratio; **Lasso** selects features by zeroing them.
- **A single tree** is a glass box (paths = rules); splits are chosen by **impurity reduction** (information gain / variance reduction).
- **RuleFit** mines rules from an ensemble, then Lasso-selects a short weighted list you read as a **scorecard** — accuracy vs readability is a dial you set.
- **Forests** are black boxes but emit a built-in **impurity importance** — fast, but **biased** (cardinality, correlation, train-based).
- Different methods give **different rankings** — importance is method-dependent, and *none of it is causation*.

Next: model-agnostic interpretability — **permutation importance** (fixes the cardinality & train-based biases — though correlation still bites) and **feature effects** (PDP / ICE), for *any* model.